

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-301b — Analog Digital Communication

Time: Three hours

Maximum Marks: 70

Instructions: Attempt all questions. Write your answers in your own words as far as practicable.

SECTION — A

Note: Long-answer questions to be answered in about 750 words each. Attempt any two. [15 × 2 = 30]

Question 1. Describe Analog and Digital Communication system. Discuss the fundamentals of Digital Modulation

OR

Describe 8-QAM and 16-QAM with suitable examples.

Question 2. Describe the detection of Signals in AWGN.

OR

Describe Error Correcting Code with reference to Convolutional Code.

SECTION — B

Note: Short-answer questions to be answered in about 200 words each. Attempt any six. [5 × 6 = 30]

Question 3. Describe the classification of Signals.

Question 4. Discuss PAM with Example.

Question 5. Describe Linear Block Code with example.

Question 6. What are Inter Symbol Interference?

Question 7. Describe different wireless standards.

Question 8. Discuss IEEE 802.11 a and n.

Question 9. Describe the classification of system.

Question 10. Discuss Hamming Code.

SECTION — C

Note: Answer the following questions in a word/sentence each.

[10 × 1 = 10]

Question 1. Full form of AWGN ,

Question 2. Define Parity

Question 3. Is 0000, 0101, 1010 and 1111 cyclic?

Question 4. Define Network communication.

Question 5. What is the function of Input transducer?

Question 6. Define BIBO stability criterion.

Question 7. Define Phase Modulation.

Question 8. Define Pulse time modulation.

Question 9. Define Phasor Diagram.

Question 10. Define Constellation Diagram.